



Carole-Jean Wu is a Director of AI Research at Meta, where she leads the Systems and Machine Learning Research team. She is a founding member and a Vice President of MLCommons – a non-profit organization that aims to accelerate machine learning innovations for the benefits of all. Dr. Wu also serves on the MLCommons Board as a Director, chaired the MLPerf Recommendation Benchmark Advisory Board, and co-chaired for MLPerf Inference. Prior to Meta/Facebook, Dr. Wu was a professor with tenure at ASU. She earned her M.A. and Ph.D. from Princeton University and B.Sc. from Cornell University.

Dr. Wu's expertise sits at the intersection of *computer architecture* and *machine learning*. Her work spans across datacenter infrastructures and edge systems with a focus on performance, energy efficiency and sustainability. She is passionate about pathfinding and tackling system challenges to enable efficient, scalable, and environmentally-sustainable AI technologies.

Dr. Wu's work has been recognized with several awards, including IEEE Micro Top Picks and ACM / IEEE Best Paper Awards. She is the recipient of NSF CAREER Award, CRA-WP Anita Borg Early Career Award Distinction of Honorable Mention, IEEE Young Engineer of the Year Award, Science Foundation Arizona Bisgrove Early Career Scholarship, and Facebook AI Infrastructure Mentorship Award. She is in the Hall of Fame of ISCA, HPCA and IISWC. Dr. Wu was the Program Co-Chair of the Conference on Machine Learning and Systems (MLSys 2022), the Program Chair of the IEEE International Symposium on Workload Characterization (IISWC 2018), and the Editor for the IEEE MICRO Special Issue on Environmentally Sustainable Computing. She currently serves on the ACM SIGARCH/SIGMICRO CARES committee, as well as the National Academies of Sciences, Engineering, Medicine workshop planning committee.

[\[Google Scholar\]](#) [\[dblp\]](#)



My work sits in the intersection of *computer architecture* and *machine learning* with the following emphasis:

- System design and optimization for deep learning
- Learning-based approaches for system design and optimization
- High-performance and energy-efficient heterogeneous CPU+GPU systems
- Performance quality modeling and energy efficiency optimization for mobile systems
- Memory system optimization
- Sustainable computing via carbon-efficient system design and management, energy harvesting and temperature-aware management for portable electronics

My work has been featured at National Academy of Engineering's *The Bridge*, the Computer Architecture Podcast Ep16: [Sustainability in a Post-AI World](#), Tech @ Meta on [Understanding computing's carbon footprint](#) and [Designing low-carbon computers](#), the Nature Magazine: [Light bulbs have energy ratings – so why can't AI chatbots?](#), and by Bloomberg Green, the Atlantic, and the HiPEAC blog: [To minimize computing's carbon footprint, the first step is to quantify lifecycle emissions](#), as well as by Stanford's MLSys seminar: [Designing AI Systems for Recommender Systems and Beyond](#), by MLPerf: [Inference v0.5 launch results](#) and [MaskRCNN2Go for MLPerf](#).

If you are interested in learning more about *Designing Computer Systems for Sustainability*, check out my course offered at *HiPEAC's Summer School*. The course includes

1. Understanding the Lay of the Land: Computing's Environmental Footprint;
2. Carbon Impact of AI;
3. Carbon Modeling and Design Optimization;
4. (Carbon)-efficient Edge Computing;
5. Carbon Optimization At-Scale: Carbon-Aware Datacenter Computing.

And, check out [Socio-Technological Challenges and Opportunities: Paths Forward](#) from ISCA-2021 Panel - The Microprocessor at 50: Societal Challenge, [Sustainable AI: Environmental Implications, Challenges and Opportunities](#) from MLSys-2022, [Designing Computer Systems for Sustainability](#) from ISCA-2024's Panel, and my thoughts on inclusive approaches to technological innovations: [Think Globally, Design Deliberately: Taking an Inclusive Approach to Innovation](#).

Recent Publications



[National Academy of Engineering *The Bridge* Winter Edition] [Scaling AI Sustainably](#)

Carole-Jean Wu, Bilge Acun, Ramya Raghavendra, Kim Hazelwood.

[HPCA-2025] [Revisiting Reliability in Large-Scale Machine Learning Research Clusters](#)

Apostolos Kokolis, Michael Kuchnik, John Hoffman, Adithya Kumar, Parth Malani, Faye Ma, Zachary DeVito, Shubho Sengupta, Kalyan Saladi, Carole-Jean Wu.

[HPCA-2025] **CORDOBA: Carbon-Efficient Optimization Framework for Computer Systems**

Mariam Elgamal, Doug Carmean, Elnaz Ansari, Okay Zed, Ramesh Peri, Srilatha Manne, Udit Gupta, Gu-Yeon Wei, David Brooks, Gage Hills, Carole-Jean Wu.

Best Paper Award Honorable Mention

[Early Version](#)

[ACM SIGARCH Computer Architecture Today] [Designing Computer Systems for Sustainability](#)

Carole Jean-Wu, Tamar Eilam, Babak Falsafi, Gage Hills, Srilatha Manne.

[Nature Magazine] [Light bulbs have energy ratings – so why can't AI chatbots?](#)

Sasha Luccioni, Boris Gamazaychikov, Sara Hooker, Regis Pierrard, Emma Strubell, Yacine Jernite, Carole-Jean Wu.

[IEEE Micro-2024] [Beyond Efficiency: Scaling AI Sustainably](#)

Carole-Jean Wu, Bilge Acun, Ramya Raghavendra, Kim Hazelwood.

Special Issue: [The Past, Present, and Future of Warehouse-Scale Computing](#)

Honors and Awards



2025 [CORDOBA: Carbon-Efficient Optimization Framework for Computer Systems](#) selected for the [IEEE HPCA Best Paper Award Honorable Mention](#)

2025 [MAD Max Beyond Single-Node: Enabling Large Machine Learning Model Acceleration on Distributed Systems](#) selected for [IEEE Micro Top Picks Honorable Mention](#)

2024 [Carbon Explorer: A Holistic Approach for Designing Carbon Aware Datacenters](#) selected for [IEEE Micro Top Picks Honorable Mention](#)

2024 [MP-Rec: Hardware-Software Co-Design to Enable Multi-Path Recommendation](#) selected for [IEEE Micro Top Picks Honorable Mention](#)

2023 2020 ISCA Paper Selected for Inclusion in *ISCA@50 25-Year Retrospective: 1996-2020: MLPerf Inference Benchmark* (Retrospective: [MLPerf](#))

One of the 98 papers (out of 1077) selected as one of the most significant and exciting papers from the ACM/IEEE International Symposium on Computer Architecture from 1996 -- 2020.

2023 [ACT: Designing Sustainable Computer Systems with an Architectural Carbon Modeling Tool](#) selected for [IEEE Micro Top Picks](#)

2022 [Towards Fair Federated Recommendation Learning: Characterizing the Inter-Dependence of System and Data Heterogeneity](#) selected for ACM Conference on Recommender Systems [Best Paper Award Finalist](#)

2022 [Chasing Carbon: The Elusive Environmental Footprint of Computing](#) selected for [IEEE Micro Top Picks](#)

2021 [MLPerf Inference Benchmark](#) selected for [IEEE Micro Top Picks](#) (Article: [The Vision Behind MLPerf: Understanding AI Inference Performance](#))

2021 [DeepRecSys: A System for Optimizing End-to-end At-scale Neural Recommendation Inference](#) selected for [IEEE Micro Top Picks Honorable Mention](#)

2020 Distinction of Honorable Mention of the CRA Anita Borg Early Career Award

2019 Facebook AI Infrastructure Mentorship Award

2019 [Genetic Improvement for GPU Code](#) selected for [ACM/IEEE ICSE Genetic Improvement on Software Best Paper Award](#)

2018 [Designing a Temperature Model to Understand the Thermal Challenges of Portable](#)





2017 NSF CAREER Award

2017 IEEE Young Engineer of the Year Award

2015 Architectural Thermal Energy Harvesting Opportunities for Sustainable Computing selected for IEEE Best of Computer Architecture Letters

2013 SFAz Bisgrove CAREER Award

2011 Intel PhD Fellowship

2011 Characterization and Dynamic Mitigation of Intra-Application Cache Interference nominated for IEEE ISPASS Best Paper Nomination

2009 Princeton Excellence in Leadership Award

Professional Service



ACM SIGARCH/SIGMICRO CARES Committee, 2024 - present.

Journal Editor

- IEEE Micro Magazine [Special Issue on Environmentally Sustainable Computing](#), 2022-23.

Executive Committee

- IEEE Technical Committee on Computer Architecture (TCCA), 2017-18.

Steering Committee

- HotCarbon: Workshop on Sustainable Computer Systems, 2023 - present.
- IEEE Intl. Symp. on Performance Analysis of Systems and Software (ISPASS), 2018-23.
- IEEE Intl. Symp. on Workload Characterization (IISWC), 2018-23.

Award Selection Committee

- IEEE TCCA Young Computer Architect Award, 2021-23.

Technical Program Chair

- Conference on Machine Learning and Systems (MLSys), 2022.
- IEEE Intl. Symp. on Workload Characterization (IISWC), 2018.

Technical Program Committee

- HPCA 2020; 2014-17; HPCA Industry Track 2025
- HotCarbon 2022, 2023
- ISCA 2014-21; ISCA Industry Track 2020
- MLSys 2020-21
- IISWC 2019; 2013-17
- MICRO 2014; 2016-17

Journal Editorial Board

- IEEE Micro 2019-23
- IEEE Computer Architecture Letters 2019-22

CRA-Widening Participation (WP) Career Mentoring Workshop

- [Finding a Research Topic](#) with Soha Hassoun, 2023.
- [Strategies for Your Career](#) with An  Settle, 2020.

Industry Initiatives and Open Source Software



Mentorship

